

Given the provided graph and our implementation of Prim’s algorithm, running the corrected code should produce an output similar to the foll

Plaintext

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```
MST Edges: [ [ '1', '4' ],
              [ '4', '3' ],
              [ '1', '6' ],
              [ '6', '2' ],
              [ '6', '5' ] ]
```

Here’s the step-by-step reasoning:

- 1. **Initialization:**
 - We start with vertex "1" (index 0) in the MST.
 - The `inMST` array is initialized as `[true, false, false, false, false, false]`.
- 2. **First Iteration:**
 - The algorithm scans all edges from vertex "1".
 - It finds the smallest edge from "1" is to "4" with a weight of 2.8.
 - Edge `["1", "4"]` is added, and `inMST` is updated to `[true, false, false, true, false, false]`.
- 3. **Second Iteration:**
 - Now vertices "1" and "4" are in the MST.
 - The algorithm finds the smallest edge connecting the MST to an outside vertex is from "4" to "3" (weight 3.4).
 - Edge `["4", "3"]` is added, updating `inMST` to `[true, false, true, true, false, false]`.
- 4. **Third Iteration:**
 - With vertices "1", "4", and "3" included, the algorithm finds the next best edge is from "1" to "6" (weight 3.6).
 - Edge `["1", "6"]` is added. Now, `inMST` becomes `[true, false, true, true, false, true]`.
- 5. **Fourth Iteration:**
 - The MST now includes "1", "4", "3", and "6".
 - It then selects the edge from "6" to "2" with a weight of 3.2.
 - Edge `["6", "2"]` is added, updating `inMST` to `[true, true, true, true, false, true]`.
- 6. **Fifth Iteration:**
 - The only remaining vertex is "5".
 - The algorithm identifies the best edge to "5" is from "6" to "5" (weight 4.6).
 - Edge `["6", "5"]` is added, and `inMST` becomes `[true, true, true, true, true, true]`.

After these 5 iterations (since a spanning tree for a graph with 6 vertices will have 5 edges), the final MST is stored in the `mstEdges` array. The

Additional Considerations:

- The ordering of the edges may vary slightly depending on how equal edges are encountered, but with the given weights, the above sequen
- This output represents the vertices (using their labels) that are connected in the Minimum Spanning Tree as determined by Prim’s algorithm